WIN TIME. FEEL COMFORT

Smart DryBag

Portable Towel Drying Bag

A consumer product concept designed to solve a common hygiene problem faced by gym users, athletes, and active individuals in everyday life.

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THE CHALLENGE



LOGO DESCRIPTION

Multiple layers of rounded rectancles means towels in different colors. There is an fan logo adds extra meaning to my logo. I used three colors mainly: blue means comfort, green means healthy, circle shaped fan is kind of clock, while white is tidyness. T letter at center is for T in Towel and Time and Tidyness

THE CHALLENGE

Problem & Solution

The Problem

Wet towels stored after gym or sports activities remain damp for extended periods, creating conditions that promote bacterial growth and unpleasant odors. Currently, there is no convenient portable solution that allows users to dry towels on the go.

- Wet towels stay damp for hours inside gym bags
- Moisture leads to rapid bacterial growth and bad odor
- No practical portable drying product exists on the market

The Solution

Smart DryBag is a portable, self-contained drying bag that uses built-in airflow, mild heat, and optional UV-C sanitation to remove moisture from towels. It is designed to be used immediately after gym sessions, keeping items hygienic and odor-free without requiring any external power source or drying equipment.

- Removes moisture through active airflow and low heat
- Optional UV-C light reduces bacteria on fabric
- Operates on a rechargeable battery — fully portable

PRODUCT OVERVIEW

Product Description & Features

Smart DryBag is a waterproof portable bag with a sealed internal compartment that actively dries wet towels through integrated technology. It is designed to be lightweight, easy to carry, and suitable for everyday gym use.

Sealed Waterproof Compartment

A waterproof internal chamber keeps wet towels contained and isolated from other belongings, preventing moisture transfer.

Integrated Mini Fan & Mild Heat

A built-in fan and low-heat element accelerate moisture evaporation inside the bag, reducing drying time significantly.

Optional UV-C Sanitation

An optional UV-C light module reduces bacteria on fabric surfaces, improving hygiene beyond simple drying.

Rechargeable & Portable

Powered by a rechargeable battery, the bag requires no external power source and fits comfortably in any gym bag or backpack.

The product accommodates different towel sizes and is designed to keep wet items fully separated from dry belongings throughout use.

FUNCTIONALITY

How It Works & Dryness Logic

Step-by-Step Process

01

Place & Seal

User places the wet towel inside the bag and closes the sealed compartment securely.

02

Humidity Detection

An internal humidity sensor measures the moisture level inside the compartment.

03

Active Drying

The system activates the fan and low-heat element to circulate air and reduce moisture from the fabric.

04

Auto-Stop

The system automatically shuts off once the towel reaches a safe, dry humidity threshold.

Humidity Threshold Logic

The drying cycle is governed by real-time humidity readings from the internal sensor. The system uses three defined states to determine operation:

Above 70%

Towel is considered wet — full drying cycle is active.

40–60%

Drying phase — airflow continues until threshold is met.

Below 40%

Towel is dry and safe to store — system stops automatically.

MARKET RESEARCH

Market Research & Target Audience

Primary research indicates strong unmet demand for portable hygiene solutions among urban gym users. Environmental factors such as humid climates further increase the relevance of the product, while growing hygiene awareness creates a receptive consumer market.

78%

Carry Wet Towels

of survey respondents carry wet towels inside their bags after gym sessions

64%

Experience Odor

reported experiencing unpleasant odor from stored wet towels

71%

Interest in Solution

expressed interest in a portable, convenient drying solution

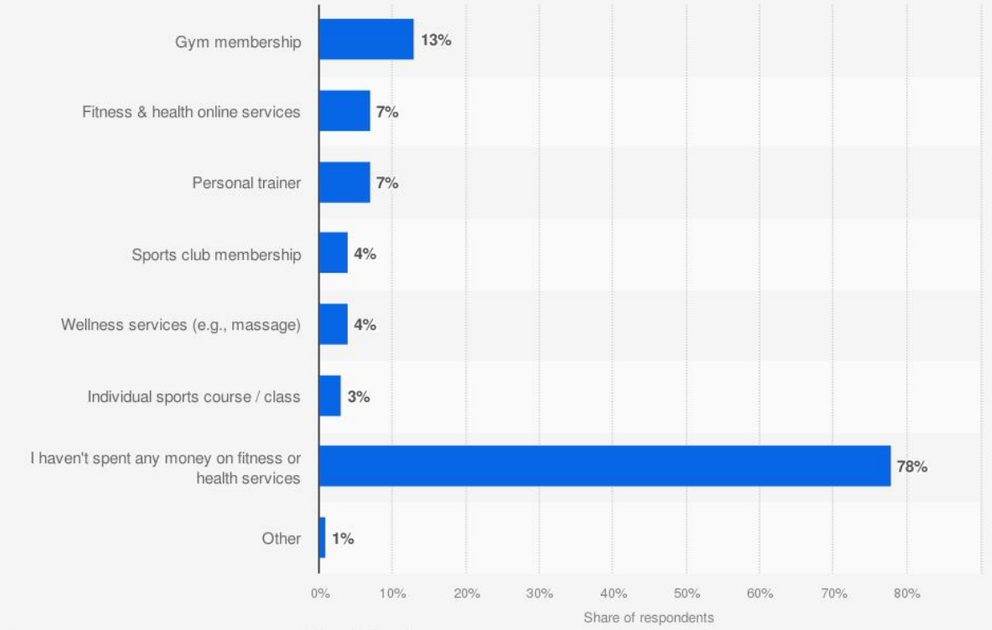
Primary Target Market

- Age 18–35, urban environment
- Students and young professionals with regular gym habits
- Health-conscious and hygiene-aware consumers

Market Context

- High volume of gym users in urban areas globally
- Humid climates increase moisture retention in fabrics
- Post-pandemic consumer sensitivity to hygiene and sanitation

Fitness and health service purchases in South Korea as of June 2025

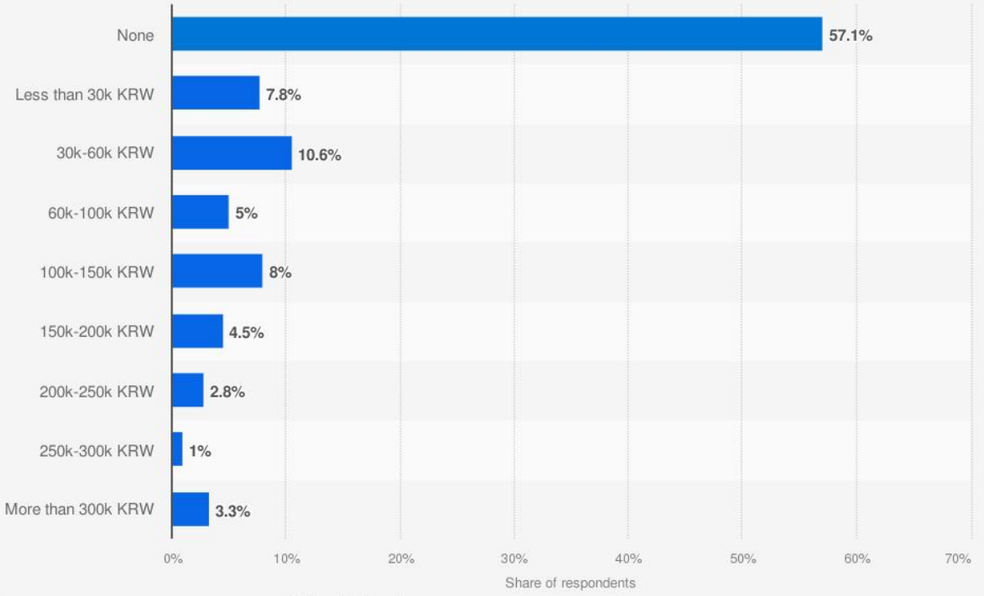


Source
Statista Consumer Insights
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Additional Information:
South Korea; July 2024 to June 2025; 2,748 respondents; 18-64 years

statista

Average monthly spending on doing sports in South Korea in 2024 (in 1,000 South Korean won)

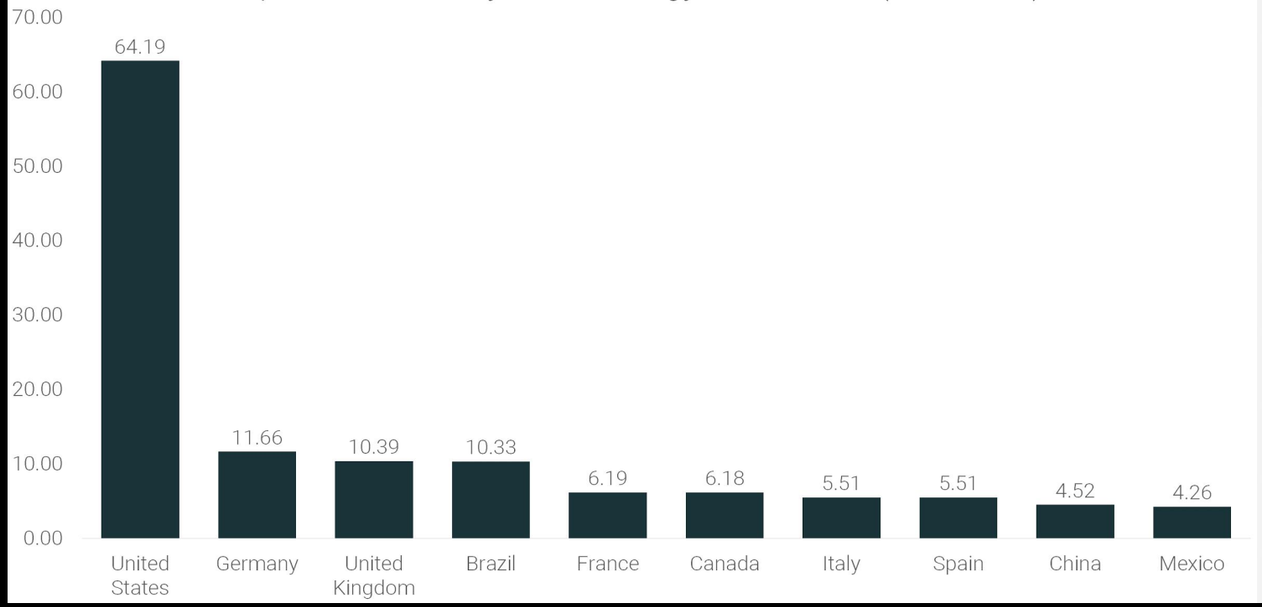


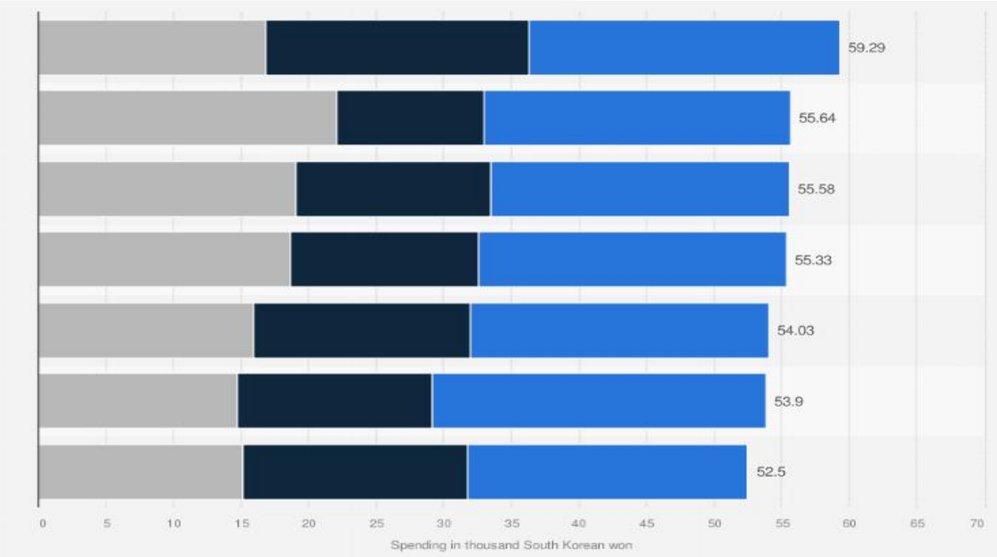
Sources
MCST (South Korea); Korea Sports Promotion Foundation; VivaCompany
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Additional Information:
South Korea; Korea Sports Promotion Foundation; VivaCompany; September 1 to November 4, 2024; 9,000 respondents; 10 years and older; Questionnaire and personal interview

statista

Top 10 countries by number of gym members (in millions)





CONCLUSION

Advantages & Conclusion

Portable & Easy to Use

The bag is designed for on-the-go use with no setup required. Users simply place the towel inside, seal the bag, and let the system operate automatically.

Superior Hygiene

Active drying and optional UV-C sanitation provide a significantly more hygienic storage method compared to placing wet towels loosely inside a gym bag.

Cost-Efficient Design

The product uses accessible and affordable components — a small fan, a low-wattage heating element, a humidity sensor, and a rechargeable battery — keeping production costs manageable.

Summary

Smart DryBag addresses a practical and common problem faced by a clearly defined consumer segment. Survey data confirms both the presence of the problem and user openness to a solution. The concept is built on simple, scalable technology and is well positioned for development as a consumer product (P2C). Further steps would include prototyping, cost analysis, and expanded market validation.